

ECEN-662: Homework 2

Due on February 6, 2026 at 11:59 PM

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Problem 1

Let

$$f_{\theta}(x) = h(x) \exp(\eta^t T - A(\eta))$$

be an s -parameter exponential family of distributions. Show that if there exists

$$\{\theta_0, \theta_1, \dots, \theta_s\} \subset \Theta$$

such that the vectors

$$(\eta(\theta_i) - \eta(\theta_0) : i \in [s])$$

are linearly independent in $\mathcal{R}^s$, then T is a minimal sufficient statistic.

Problem 2

Suppose that T_1 is sufficient and T_2 is minimally sufficient. Let U be an unbiased estimator of θ and defined

$$U_1 = \mathbb{E}_\theta[U|T_1] \quad \text{and} \quad U_2 = \mathbb{E}_\theta[U|T_2]$$

Show that $U_2 = \mathbb{E}_\theta[U_1|T_2]$ and hence $\text{Var}_\theta[U_2] \leq \text{Var}_\theta[U_1]$.

Problem 3**Problem 4**